HIGHWAY ENGINEERING LAB

Course Code 23A01602P | III B.Tech - II Semester | JNTUA R23

Course Details

Programme	B.Tech. Civil Engineering
Regulation	JNTUA R23
Course	Highway Engineering Lab (Practical)
Course code	23A01602P
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Experiments	Part I: Tests on Aggregates (7) Part II: Tests on Bitumen (7)
Text book	Highway Material Testing Manual, Khanna, Justo and Veera Raghavan, Nemchand Brothers
Reference books	IS 383:1993; IS 1201-1220 (1978); IRC SP 53-2010; MS-2 Manual (Marshall Mix Design, 2002)
Online resource	https://ts-nitk.vlabs.ac.in/

Course objectives

- Understand the properties and behaviour of aggregates and bitumen used in highway construction.
- Perform standard laboratory tests on aggregates and bitumen to judge their suitability for road construction.
- Analyse the strength, durability and performance characteristics of pavement materials.
- Assess the quality and compliance of highway materials with standard specifications.
- Develop hands-on skills for material testing and interpretation of test results.

Course outcomes

1. Determine the physical properties of coarse aggregates — specific gravity, water absorption and shape.
2. Evaluate the mechanical properties — abrasion resistance, impact strength and crushing value.
3. Analyse the physical and chemical properties of bituminous materials through standard tests.
4. Perform Marshall stability tests and assess the optimum binder content of bituminous mixes.
5. Interpret test results to judge the suitability of aggregates and bitumen for pavement construction.

About this record

For every experiment the **inputs** (observations / readings taken in the lab) and the **outputs** (calculated results and comparison with the specification) are given. The numbers shown are **sample readings** worked out to illustrate the method — replace them with the readings from your own test and recalculate. Limits quoted are typical values from IS / IRC / MoRTH; check the current edition of the code or your lab manual.

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General precautions	Wear apron, gloves and safety glasses; use tongs for hot moulds; keep bitumen away from open flame; weigh to the accuracy asked (0.1 g / 1 g); record temperature of test; clean all apparatus after use.

PART I - TESTS ON AGGREGATES

Experiment 1: Specific Gravity of Coarse Aggregate

Aim

To determine the specific gravity (and apparent specific gravity) of a coarse aggregate sample. Reference: IS 2386 (Part III).

Apparatus

- Pycnometer (1 litre) with brass cap, balance (0.5 g accuracy), oven (100 – 110 °C), sieves (10 mm, 4.75 mm), tray, dry cloth, distilled water.

Theory

- Specific gravity G = weight of aggregate ÷ weight of an equal volume of water. It decides the **quality (strength) of the stone** : good road aggregates have G = 2.6 – 2.9 and low porosity.

Procedure

1. Take about 500 g of aggregate passing 10 mm and retained on 4.75 mm; wash, soak in water for 24 hours and bring it to the saturated surface-dry (SSD) condition with a cloth.
2. Weigh the empty dry pycnometer (W_1).
3. Put the SSD aggregate (about 500 g) in the pycnometer and weigh (W_2).
4. Fill with water to the top (remove air by rolling and tapping), put the cap on and weigh (W_3).
5. Empty, clean and fill the pycnometer with water only and weigh (W_4).
6. Dry the aggregate in the oven for 24 hours, cool and weigh it (W_{dry}) to get the apparent specific gravity.

Observations (inputs)

Observation	Symbol	Weight (g)
Empty pycnometer	W_1	650
Pycnometer + SSD aggregate	W_2	1150
Pycnometer + aggregate + water	W_3	1830
Pycnometer + water only	W_4	1520
Oven-dry weight of aggregate	W_{dry}	494

Calculations (output)

- Weight of SSD aggregate = $W_2 - W_1 = 1150 - 650 =$ **500 g** .
- Specific gravity $G = (W_2 - W_1) / [\,(W_2 - W_1) - (W_3 - W_4)\,]$ = 500 / [500 – (1830 – 1520)] = 500 / 190 = **2.63** .
- Apparent specific gravity = $W_{dry} / [\,W_{dry} - (W_3 - W_4)\,]$ = 494 / (494 – 310) = 494 / 184 = **2.68** .

- Water absorption = (500 − 494)/494 × 100 = **1.21 %** (used in Experiment 7).

Result

Specific gravity of the coarse aggregate = **2.63** ; apparent specific gravity = **2.68** (within the usual range 2.5 – 3.0 → suitable).

Experiment 2: Abrasion Value (Los Angeles) of Coarse Aggregate

Aim

To find the Los Angeles abrasion value of the aggregate to judge its resistance to wear (hardness). Reference: IS 2386 (Part IV).

Apparatus

- Los Angeles abrasion machine (hollow steel cylinder 700 mm dia × 500 mm long, rotated at 20 – 33 rpm, with a shelf inside), 11 steel balls (about 48 mm dia, total 4584 g) for Grading B, IS sieve 1.70 mm, balance, oven, tray.

Procedure

1. Take 5 kg of dry aggregate of Grading B: 2.5 kg passing 20 mm / retained 12.5 mm and 2.5 kg passing 12.5 mm / retained 10 mm.
2. Place the sample and the abrasive charge (11 steel balls) in the cylinder and close the cover.
3. Rotate the machine for **500 revolutions** at 30 – 33 rpm.
4. Remove the material, pass it through a **1.70 mm** sieve (wash and dry the retained part).
5. Weigh the aggregate retained on the 1.70 mm sieve (W_2).

Observations (inputs)

Observation	Value
Grading used / number of balls	B / 11
Revolutions at 30 – 33 rpm	500
Original weight of sample, W_1	5000 g
Weight retained on 1.70 mm sieve, W_2	3650 g

Calculations (output)

- Weight passing 1.70 mm = 5000 − 3650 = 1350 g.
- **Los Angeles abrasion value = $(W_1 - W_2)/W_1 \times 100$** = (5000 − 3650) / 5000 × 100 = **27 %** .

Pavement layer (typical limit, max %)	LA abrasion value
Bituminous concrete (wearing course)	30
Dense bituminous macadam / bituminous macadam	35
WMM / WBM surface course	40
WBM base and sub-base	50 and above (as per code)

Result

Los Angeles abrasion value = **27 %** , which is below 30 %, so the aggregate is **suitable** even for bituminous concrete wearing course.

- Precaution: the sample must be oven-dry; clean the balls; the cover and the shelf should be tightly fixed.

Experiment 3: Impact Value of Coarse Aggregate

Aim

To determine the aggregate impact value, which measures the toughness (resistance to sudden shock or impact). Reference: IS 2386 (Part IV).

Apparatus

- Impact testing machine with a hammer of 13.5 – 14 kg falling freely through **380 mm** , steel cup (102 mm dia, 50 mm deep), cylindrical measure (75 mm dia, 50 mm deep), tamping rod (10 mm dia, 230 mm long), **2.36 mm** and 12.5/10 mm IS sieves, balance (0.1 g), oven.

Procedure

1. Take aggregate passing 12.5 mm and retained on 10 mm, dry it in the oven (100 – 110 °C, 4 hours) and cool.
2. Fill the cylindrical measure in **three layers** , tamping each layer 25 times; strike off the top; weigh the aggregate (W_1).
3. Transfer the sample to the steel cup of the machine and tamp it 25 times.
4. Raise the hammer to 380 mm and let it fall freely — **15 blows** , at an interval of at least 1 second.
5. Remove the crushed aggregate and sieve it on the **2.36 mm** IS sieve; weigh the fraction passing (W_2) and the fraction retained.
6. Repeat with a second sample and take the mean.

Observations (inputs)

Observation	Test 1	Test 2
Weight of aggregate in the cup, W_1 (g)	352	348
Weight passing 2.36 mm sieve, W_2 (g)	76	74
Weight retained on 2.36 mm sieve (g)	276	274

Calculations (output)

- **Aggregate impact value (AIV) = $W_2 / W_1 \times 100$**
- Test 1 = 76 / 352 × 100 = 21.59 %; Test 2 = 74 / 348 × 100 = 21.26 %
- Mean AIV = (21.59 + 21.26) / 2 = **21.4 %** .
- Check: W_2 + retained = 352 (weight loss during sieving must be less than 1 g).

Use of aggregate (typical limit, max %)	AIV
Wearing (surface) course	30
Bituminous macadam / base course	35 - 40
WBM sub-base	45 - 50

- Classification: 10 % exceptionally strong; 10 – 20 % strong; 20 – 30 % satisfactory for road surfacing; > 35 % weak.

Result

Aggregate impact value = **21.4 %** (< 30 %) → the aggregate is **strong and suitable** for wearing course.

Experiment 4: Elongation Index of Coarse Aggregate

Aim

To find the elongation index of the aggregate — the percentage (by weight) of particles whose greatest dimension is more than 1.8 times the mean size. Reference: IS 2386 (Part I).

Apparatus

- Length gauge, set of IS sieves (50, 40, 25, 20, 16, 12.5, 10, 6.3 mm), balance, tray.

Procedure

1. Take a representative sample (at least 200 pieces of each fraction) and sieve it into the standard size ranges.
2. Weigh each fraction that has at least 5 % of the total (W).
3. Try each particle of a fraction in the **slot of the length gauge** for that size; the particles that **cannot pass** through the slot are **elongated** .
4. Weigh the elongated pieces of each fraction (e).
5. Elongation index = total weight of elongated particles as a percentage of the total weight of the sample.

Observations (inputs) — see the combined table in Experiment 5

Result of calculations (output)

- Total weight of the sample = **5900 g** ; total weight of elongated particles = **970 g** .
- **Elongation index = (Σ e / Σ W) × 100** = 970 / 5900 × 100 = **16.4 %** .

Result

Elongation index = **16.4 %** . (The limit for the combined flakiness + elongation index is 35 % for bituminous mixes — see Experiment 5.)

Experiment 5: Flakiness Index of Coarse Aggregate

Aim

To determine the flakiness index — the percentage (by weight) of particles whose least dimension (thickness) is less than 0.6 times the mean size. Reference: IS 2386 (Part I).

Apparatus

- Thickness gauge (metal plate with slots), IS sieves (63, 50, 40, 25, 20, 16, 12.5, 10, 6.3 mm), balance (accuracy 0.1 % of weight), tray.

Procedure

1. Sieve a representative sample (minimum quantity as per IS) into the size fractions; weigh each fraction having more than 5 % of the total.
2. Pass the particles of each fraction through the corresponding slot of the **thickness gauge**; particles that **pass** the slot are **flaky**.
3. Weigh the flaky particles of each fraction (w).
4. Flakiness index = total weight passing the gauge ÷ total weight of the sample × 100.

Observations (inputs) — combined table for Experiments 4 and 5

Size (passing - retained), mm	Thickness gauge (mm)	Length gauge (mm)	Weight W (g)	Flaky w (g)	Elongated e (g)
40 - 25	19.50	58.50	1200	150	190
25 - 20	13.50	40.50	1500	200	250
20 - 16	10.80	32.40	1100	160	190
16 - 12.5	8.55	25.65	900	130	150
12.5 - 10	6.75	20.25	700	100	120
10 - 6.3	4.89	14.67	500	80	70
Total			**5900**	**820**	**970**

- Thickness gauge = 0.6 × mean size; length gauge = 1.8 × mean size (e.g. 40 - 25: mean 32.5 → 19.5 and 58.5).

Calculations (output)

- **Flakiness index** $= (\Sigma w / \Sigma W) \times 100 = 820 / 5900 \times 100 =$ **13.9 %**.
- Elongation index $= 970 / 5900 \times 100 =$ **16.4 %**.
- **Combined index** $= 13.9 + 16.4 =$ **30.3 %**.

Limit (typical, IRC / MoRTH)	Value
Flakiness index for WBM / base courses	15 % max
Combined flakiness and elongation (bituminous mixes)	35 % max

Limit (typical, IRC / MoRTH)	Value
Combined index (WMM / granular layers)	30 % max

Result

Flakiness index = **13.9 %** , elongation index = **16.4 %** , combined = **30.3 %** → within the limit for bituminous mixes (35 %); aggregate shape is **acceptable** . Flaky and elongated particles are weak and give poor interlock and high voids.

Experiment 6: Aggregate Crushing Value of Coarse Aggregate

Aim

To determine the aggregate crushing value (ACV), which measures the strength of the aggregate against gradually applied compressive load. Reference: IS 2386 (Part IV).

Apparatus

- Cylindrical steel mould (152 mm dia) with plunger and base plate, compression testing machine (capacity 40 tonnes, loading at 4 t per minute), tamping rod, balance, **2.36 mm** and 12.5 / 10 mm IS sieves, measure (115 mm dia, 180 mm high).

Procedure

1. Take aggregate passing 12.5 mm and retained on 10 mm (dry it in the oven, then cool); fill the cylindrical measure in **three layers** , tamping each layer 25 times; weigh the sample (W_1).
2. Put the sample into the mould in three layers (each tamped 25 times), level the surface and place the plunger.
3. Place the mould on the compression testing machine and apply a load of **40 tonnes (400 kN) at a uniform rate in 10 minutes** .
4. Release the load, remove the crushed material and sieve it through the **2.36 mm** IS sieve.
5. Weigh the fraction passing the sieve (W_2). Repeat with a second sample.

Observations (inputs)

Observation	Test 1	Test 2
Weight of sample, W_1 (g)	3010	2995
Weight passing 2.36 mm sieve, W_2 (g)	845	838
Load applied / time	400 kN / 10 min	400 kN / 10 min

Calculations (output)

- **Aggregate crushing value = $W_2 / W_1 \times 100$**
- Test 1 = 845 / 3010 × 100 = 28.07 %; Test 2 = 838 / 2995 × 100 = 27.98 %
- Mean ACV = **28.0 %** .

Use (IS 383 / IRC limit, max)	ACV
Concrete wearing surfaces	30 %
Concrete other than wearing surfaces	45 %
Base course of roads	45 %

Result

Aggregate crushing value = **28.0 %** (< 30 %) → the aggregate is **suitable for wearing surface** of roads and for concrete.

Experiment 7: Water Absorption Capacity of Coarse Aggregate

Aim

To determine the water absorption of the coarse aggregate; it indicates porosity and durability. Reference: IS 2386 (Part III).

Apparatus

- Balance (accuracy 0.5 g), oven (100 – 110 °C), wire basket, container for water, dry absorbent cloth, tray, IS sieve 10 mm.

Procedure

1. Take about 2 kg of aggregate retained on the 10 mm sieve, wash it to remove fines and dust.
2. Immerse in water at 22 – 32 °C for **24 ± 0.5 hours** ; shake to remove entrapped air.
3. Remove the aggregate, roll it in a dry cloth until the visible films of water disappear (saturated surface-dry); weigh (B).
4. Dry the aggregate in the oven at 100 – 110 °C for 24 hours, cool in an air-tight container and weigh (C).

Observations (inputs)

Observation	Symbol	Weight (g)
Saturated surface-dry weight	B	2000
Oven-dry weight	C	1976

Calculations (output)

- Water absorbed = B − C = 2000 − 1976 = 24 g.
- **Water absorption (%) = (B − C) / C × 100** = 24 / 1976 × 100 = **1.21 %** .

Reference	Limit
Aggregates for bituminous surfacing and road layers (MoRTH)	2 % max
Good road aggregate	usually 0.1 – 2 %

Result

Water absorption = **1.21 %** (< 2 %) → the aggregate has low porosity and is **acceptable** ; it matches the value found in Experiment 1.

PART II - TESTS ON BITUMEN

Experiment 8: Specific Gravity of Bitumen / Asphalt

Aim

To determine the specific gravity of a bitumen sample using a pycnometer. Reference: IS 1202.

Apparatus

- Specific gravity bottle (pycnometer) 50 ml with stopper, balance (0.01 g accuracy), water bath at 27 °C, distilled water, spatula.

Procedure

1. Clean and dry the bottle and weigh it with the stopper (W_1).
2. Fill it with distilled water, keep in the water bath at 27 °C for 30 minutes, dry the outside and weigh (W_2).
3. Empty and dry the bottle; pour **half-filled (about 20 g)** heated bitumen (free of air bubbles), cool and weigh (W_3).
4. Fill the remaining space with distilled water, replace the stopper, keep in the water bath for 30 minutes at 27 °C, wipe and weigh (W_4).

Observations (inputs)

Observation	Symbol	Weight (g)
Empty bottle	W_1	25.20
Bottle + water	W_2	75.20
Bottle + bitumen	W_3	45.20
Bottle + bitumen + water	W_4	75.60

Calculations (output)

- **Specific gravity $G = (W_3 - W_1) / [(W_2 - W_1) - (W_4 - W_3)]$**
- $= (45.20 - 25.20) / [(75.20 - 25.20) - (75.60 - 45.20)] = 20.00 / (50.00 - 30.40) = 20.00 / 19.60 =$ **1.02** .

Result

Specific gravity of the bitumen at 27 °C = **1.02** (paving bitumen: 0.97 – 1.02, VG grades min 0.99) → **satisfactory** .

- Use: to convert bitumen content by weight to volume and for calculating voids in Marshall mix design.

Experiment 9: Penetration Test on Bitumen

Aim

To determine the hardness (consistency) of bitumen by measuring the depth to which a standard needle penetrates in 5 seconds under a 100 g load at 25 °C. Reference: IS 1203.

Apparatus

- Penetrometer with standard needle (total moving weight 100 g), sample container (55 mm dia, 35 mm deep), water bath at 25 °C (± 0.1 °C), transfer dish, stop watch, thermometer.

Procedure

1. Heat the bitumen to a pouring temperature (not more than 90 °C above its softening point), stir to remove air bubbles and pour into the container; cool in air for 1 – 1.5 hours.
2. Keep the container in the water bath at **25 °C** for 1 – 1.5 hours.
3. Place the container on the penetrometer table; bring the needle tip to just touch the surface of the sample and set the dial to zero.
4. Release the needle for **5 seconds** and read the penetration on the dial (unit = 0.1 mm).
5. Make **at least three readings** at points 10 mm or more from the edge and from each other, wiping the needle each time.

Observations (inputs)

Observation	Reading 1	Reading 2	Reading 3
Load / time / temperature	100 g / 5 s / 25 °C	100 g / 5 s / 25 °C	100 g / 5 s / 25 °C
Penetration (0.1 mm)	64	66	65

Calculations (output)

- Mean penetration = (64 + 66 + 65) / 3 = **65** (units of 0.1 mm) = 6.5 mm.
- Spread of readings = 66 − 64 = 2 (allowed for pen. 50 – 149: not more than 4 units) √.

Result

Penetration of the bitumen = **65** → it belongs to **60/70 penetration grade** (60 – 70). Grades: 30/40 hard, 60/70 and 80/100 for hot and moderate climates respectively.

Experiment 10: Viscosity of Bituminous Binder

Aim

To determine the viscosity of a bituminous binder (cut-back / tar) by the flow time of a fixed volume through a standard orifice. Reference: IS 1206.

Apparatus

- Standard Tar Viscometer (STV) with a cup, 10 mm orifice, a valve rod, receiver (graduated flask), thermostatically-controlled water bath (heated), stop watch, thermometers, sample.

Procedure

1. Clean the cup, close the orifice with the valve rod and fill the cup with the sample up to the mark (after heating to the test temperature, e.g. 40 °C).
2. Maintain the test temperature by circulating water in the jacket for at least 5 minutes after the sample has reached it.
3. Place the receiver below the orifice; lift the valve rod and **start the stop watch** .
4. Stop the watch when **50 ml** of the binder has been collected; record the time in seconds.
5. Repeat the test and take the mean.

Observations (inputs)

Observation	Trial 1	Trial 2
Orifice / temperature	10 mm / 40 °C	10 mm / 40 °C
Volume collected	50 ml	50 ml
Time of flow (s)	85	87

Calculations (output)

- **Viscosity (tar viscometer) = mean time for 50 ml to flow** = (85 + 87) / 2 = **86 seconds** (at 40 °C, 10 mm orifice).
- Higher flow time → more viscous binder. Viscosity reduces rapidly with temperature, so it is used to fix the **mixing and compaction temperatures** of a mix.

Result

Viscosity at 60 °C (poise): VG-10 = 800 – 1200, VG-20 = 1600 – 2400, VG-30 = 2400 – 3600, VG-40 = 3200 – 4800. Kinematic viscosity at 135 °C (min, cSt): 250, 300, 350 and 400 respectively.

Viscosity of the binder = **86 s** (STV, 10 mm orifice, 40 °C). The binder is suitable for the grade specified for the test temperature (compare with the cut-back grade table of IS 217 / 1206).

Experiment 11: Softening Point of Bitumen (Ring and Ball)

Aim

To determine the softening point of bitumen — the temperature at which the bitumen softens and a steel ball sinks through it by 25 mm under standard conditions. Reference: IS 1205.

Apparatus

- Brass ring (inside dia 19 mm, depth 6.4 mm), steel ball (9.5 mm dia, 3.5 g), ball guide, ring holder with a bottom plate (25 mm below the ring), 800 ml beaker, thermometer (0 – 100 °C),

distilled water (glycerine if above 80 °C), Bunsen burner with stand and wire gauze.

Procedure

1. Heat the bitumen to the pouring temperature, pour into two brass rings placed on a greased plate; level the surface by a hot knife after cooling.
2. Fill the beaker with distilled water at **5 °C** and place the ring assembly in it; keep for 15 minutes.
3. Place a steel ball on each ring, put the thermometer in position and heat the beaker so that the temperature rises at **5 ± 0.5 °C per minute** .
4. Record the temperature when the bitumen touches the bottom plate (after the ball has sunk through 25 mm).
5. Repeat with the second ring; the two values should not differ by more than 1 °C.

Observations (inputs)

Observation	Ring 1	Ring 2
Starting water temperature	5 °C	5 °C
Rate of heating	5 °C / min	5 °C / min
Temperature when the ball touches the plate	49.0 °C	49.5 °C

Calculations (output)

- **Softening point = mean of the two temperatures** = (49.0 + 49.5) / 2 = **49.3 °C** (difference 0.5 °C < 1 °C √).

Grade (IS 73)	Softening point min (°C)
80/100 penetration grade	40
60/70 penetration grade	45
VG-10 / VG-20 / VG-30 / VG-40	40 / 45 / 47 / 50

Result

Softening point of the bitumen = **49.3 °C** (> 45 °C) → **satisfactory** for 60/70 grade bitumen; a higher softening point means a binder that is less susceptible to temperature changes (suitable for hot climates).

Experiment 12: Ductility Value of Bitumen

Aim

To find the ductility of a bitumen sample — the distance in cm to which a standard briquette can be stretched before the thread breaks. Reference: IS 1208.

Apparatus

- Briquette mould (brass, with side pieces), ductility machine with a water bath (pulling rate 50 mm / minute), thermometer, knife, glycerine and dextrin (to prevent sticking), water at 27 °C.

Procedure

1. Assemble the mould on a brass plate coated with a release mixture (glycerine + dextrin); pour the heated bitumen carefully (no air bubbles) into the mould, slightly overfilling.
2. Cool in air for 30 minutes, then in a water bath at 27 °C for 30 minutes; trim the surface level by a hot knife.
3. Keep the mould in the water bath at **27 °C** for 85 – 95 minutes; remove the side pieces.
4. Hook the clips to the machine pins and pull at **50 mm / minute**, at 27 ± 0.5 °C.
5. Note the distance at which the thread of bitumen breaks. Repeat for **three** briquettes.

Observations (inputs)

Observation	Briquette 1	Briquette 2	Briquette 3
Test temperature / pulling rate	27 °C / 50 mm per min	27 °C / 50 mm per min	27 °C / 50 mm per min
Distance when the thread breaks (cm)	82	85	80

Calculations (output)

- **Ductility = mean of the three readings** = (82 + 85 + 80) / 3 = **82.3 cm**.

Specification (IS 73)	Ductility min (cm)
VG-10 and VG-30 at 25 °C	75
VG-20	50
VG-40	40
60/70 penetration grade at 27 °C	50

Result

Ductility of the bitumen = **82.3 cm** → greater than 75 cm, so the bitumen has good binding and **flexibility properties** and is suitable for the pavement. Low ductility means a brittle binder that cracks.

Experiment 13: Loss of Bitumen on Heating

Aim

To determine the loss in mass of bitumen when heated at 163 °C for 5 hours (volatile loss) — it shows the ageing and hardening tendency of the binder. Reference: IS 1212.

Apparatus

- Thin-film oven (thermostatically-controlled at 163 ± 1 °C), flat-bottom metal dish (140 mm dia, 9.5 mm deep) or 85 mm cup, balance (0.01 g), desiccator, penetrometer (for the penetration retained).

Procedure

1. Heat the bitumen to the pouring condition and pour **50 g** into the weighed dish; cool and weigh (W_1).
2. Place the dish on the rotating shelf of the oven at **163 °C** and keep it for **5 hours** .
3. Remove, cool in a desiccator and weigh again (W_2).
4. Test the residue for penetration to find the retained penetration (optional).

Observations (inputs)

Observation	Value
Initial weight of bitumen, W_1	50.00 g
Temperature / time of heating	163 °C / 5 hours
Final weight after heating, W_2	49.80 g

Calculations (output)

- Loss in weight = 50.00 − 49.80 = 0.20 g.
- **Loss on heating (%) = $(W_1 - W_2) / W_1 \times 100$** = 0.20 / 50 × 100 = **0.4 %** .

Result

Loss on heating = **0.4 %** (limit: maximum 1 % by mass) → the bitumen contains **few volatiles** , will not age or harden too quickly and is **satisfactory** .

- Optionally test the residue for penetration and compare it with the original value (retained penetration) as required by the specification.

Experiment 14: Bitumen Extraction Test

Aim

To determine the bitumen (binder) content of a bituminous paving mix by solvent extraction, and to recover the aggregate for gradation tests. Reference: ASTM D2172 / MS-2 Manual (method B – centrifuge).

Apparatus

- Centrifuge extractor (bowl with cover, filter ring, receiver for solvent), solvent (trichloroethylene or benzene), pan, balance (0.1 g), oven, dry filter ring, sieves, safety equipment (use in a fume hood).

Procedure

1. Weigh a representative sample of the loose paving mix (about 1000 g) and heat it to loosen the particles (W_1).
2. Place the sample in the bowl, cover it with solvent and let it stand for **15 minutes** .
3. Fit the filter ring, cover and start the centrifuge slowly; increase the speed until the solvent stops flowing. Collect the solvent in the receiver.
4. Add 200 ml of fresh solvent and repeat until the extract is **clear (light straw colour)** .
5. Remove the aggregate and the filter ring, dry them in the oven (110 °C) to constant weight and weigh (W_2 — aggregate with the filler retained).
6. Determine the ash in the filter and the fines washed into the solvent (correction) if required.

Observations (inputs)

Observation	Value
Weight of bituminous mix before extraction, W_1	1000.0 g
Weight of extracted aggregate and filler (after drying), W_2	944.0 g
Filter-ring ash and fines correction	included in W_2

Calculations (output)

- Weight of bitumen = $W_1 - W_2$ = 1000 − 944 = 56.0 g.
- **Bitumen content (% by weight of mix) = $(W_1 - W_2) / W_1 \times 100$** = 56 / 1000 × 100 = **5.6 %** .
- Bitumen content by weight of aggregate = 56 / 944 × 100 = 5.93 %.
- Aggregate gradation: sieve the recovered aggregate and compare with the job-mix formula.

Result

Bitumen content of the mix = **5.6 %** by weight of mix. Compare it with the design (optimum) binder content from the Marshall test; the allowed tolerance is about ± 0.3 %.

SUMMARY AND VIVA

Summary of Inputs and Outputs (sample values)

No.	Experiment	Inputs (observations)	Output (result)	Typical limit
1	Specific gravity	W_1 650, W_2 1150, W_3 1830, W_4 1520 g	G = 2.63	2.5 - 3.0
2	Los Angeles abrasion	W_1 5000 g, W_2 3650 g, 500 rev	27 %	30 % (BC)
3	Impact value	W_1 352 / 348, W_2 76 / 74 g	21.4 %	30 %
4	Elongation index	total 5900 g, elongated 970 g	16.4 %	combined 35 %
5	Flakiness index	total 5900 g, flaky 820 g	13.9 %	15 % (WBM)
6	Crushing value	W_1 3010 / 2995, W_2 845 / 838 g	28.0 %	30 %
7	Water absorption	B 2000 g, C 1976 g	1.21 %	2 %
8	Specific gravity of bitumen	W_1 25.20, W_2 75.20, W_3 45.20, W_4 75.60 g	1.02	0.99 min
9	Penetration	64, 66, 65 (0.1 mm)	65 → 60/70 grade	60 - 70
10	Viscosity (STV)	85 s, 87 s for 50 ml	86 s	per grade
11	Softening point	49.0 and 49.5 °C	49.3 °C	45 °C min
12	Ductility	82, 85, 80 cm	82.3 cm	50 - 75 cm min
13	Loss on heating	50.00 g to 49.80 g	0.4 %	1 % max
14	Bitumen extraction	mix 1000 g, aggregate 944 g	5.6 % binder	design ± 0.3 %

- All sample results satisfy the limits, so the aggregate and the bitumen are judged **suitable for pavement construction** (Course Outcome 5).

Test Standards at a Glance

Test	Standard	Key conditions
Specific gravity and water absorption (aggregate)	IS 2386 (Part III)	pycnometer; 24 h soaking; oven 100 - 110 °C
Los Angeles abrasion	IS 2386 (Part IV)	500 revolutions, 1.70 mm sieve
Impact value	IS 2386 (Part IV)	15 blows, 380 mm fall, 2.36 mm sieve
Crushing value	IS 2386 (Part IV)	40 t in 10 min, 2.36 mm sieve

Test	Standard	Key conditions
Flakiness / elongation	IS 2386 (Part I)	thickness gauge 0.6 × mean; length gauge 1.8 × mean
Specific gravity of bitumen	IS 1202	pycnometer at 27 °C
Penetration	IS 1203	100 g, 5 s, 25 °C
Softening point	IS 1205	ring and ball, 5 °C/min
Viscosity	IS 1206	tar viscometer, 50 ml flow time
Ductility	IS 1208	27 °C, 50 mm per min
Loss on heating	IS 1212	163 °C for 5 h
Extraction	ASTM D2172 / MS-2	centrifuge with solvent

Marshall Test (Course Outcome 4) — For Reference

- Not in the list of experiments above, but Course Outcome 4 refers to it. **Marshall stability** is the maximum load (kN) carried by a 101.6 mm dia × 63.5 mm high specimen loaded at 50 mm/min at 60 °C; **flow value** is the deformation (mm) at maximum load.
- Specimens are prepared at binder contents such as 4.5, 5.0, 5.5, 6.0 and 6.5 %; the optimum binder content is the average of the contents at maximum stability, maximum bulk density and 4 % air voids (MS-2).
- Typical requirements (bituminous concrete, 75 blows each face): stability min 9 kN; flow 2 - 4 mm; air voids 3 - 5 %; VMA and VFB as per grading.

Viva Questions

1. What is the specific gravity of good road aggregate and why is it needed?
2. Why is a Los Angeles machine rotated for 500 revolutions and what is the 1.70 mm sieve for?
3. What is the difference between impact value and crushing value?
4. Define flaky and elongated particles; what are the thickness and length gauges?
5. Why should water absorption of aggregate for bituminous mixes be low?
6. What does a penetration value of 65 mean? What grade is it?
7. What is softening point and why is it important in hot climates?
8. Why is ductility measured at 27 °C at 50 mm per minute?
9. What does loss on heating show about bitumen?
10. How is the binder content of a mix found by extraction and how does it relate to the Marshall test?